A Theory of Optimal Disaster Insurance under Intermediary Constraints

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Motivation

- Property insurance is central for homeowners to manage climate risks.
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  - Rate caps improve affordability but drive insurer exits (Oh et al 2021; Liu and Liu, 2024)
  - Direct subsidies likewise mute signals and could delay adaptation (Ge et al 2025)
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  - Residual insurers ensure supply, but may undercut private insurers (Abramson et al 2025).

- The literature provides reduced-form evidence of these distortions.
- We lack a structural investigation into optimal intervention.

- **RQ:** Which insurance regulation optimally balances keeping homeowners insured with insurers' long-run financial stability?

This Paper

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- Presents 3 empirical facts on how insurance contractions transmit into housing markets.
- Develops a dynamic GE model with housing and insurance intermediaries.
  - **Intertemporal trade-off**: more insurance take-up protects home values, but tightens insurer financial constraints, steepening insurance contractions upon shocks.
  - Two occasionally binding constraints (household collateral and insurer capital)
  - Solve a decentralized and constrained-efficient (planner) equilibrium + derive implicit insurance subsidies/taxes that maximize HH welfare.
  - Insurance tax or subsidy depends on future marginal propensity to insure (MPI).
  - Calibrated to the California housing market.

Main Findings

Motivating Empirical Facts:

- Insurance mitigates house price reductions upon weather shocks.
- Insurance market contractions steepen in insurer financial constraints.
- Insurance contractions negatively affect house prices conditional on weather exposure.

Model Results:

- High household leverage + low insurance generates severe drops in house prices and consumption following weather shocks, which households inefficiently do not internalize.
- A planner optimally subsidizes insurance, averaging 2.7% and ↑ welfare by +0.2% CEV.
- The optimal subsidy is dynamic: high when HHs are leveraged, low when insurers are constrained.
- Letting regulators set prices directly reduces welfare and makes insurance gaps persistent.

Literature

1. Effects of insurance market regulation on homeowners and capital markets (Koijen and Yogo, 2015; Born and Klimaszewski-Blettner, 2013; Becker and Opp, 2013; Oh et al., 2021; Liu and Liu, 2024).

   This paper: Derive the optimal time-consistent insurance policy in a dynamic housing model.

2. Demand- and supply-side frictions in insurance markets: financial constraints (Sastry et al., 2025a), tail-risk neglect (Kunreuther & Pauly, 2004), learning (Gallagher, 2014) and limited attention (Cookson et al., 2024) **(Demand)**; institutional constraints (Koijen & Yogo, 2022), regulatory frictions (Oh et al., 2021), informational frictions (Boomhower et al., 2024), capital market frictions (Eastman et al., 2024; Solomon, 2024) and coordination failures (Ibragimov et al., 2009) **(Supply)**.

   This paper: A model with occasionally binding constraints on both the household and intermediary side.

3. Role of financial intermediaries in shaping macro-economic fluctuations. (Gertler and Kiyotaki, 2010; Coimbra and Rey, 2024; Favara and Imbs, 2015; Jamilov and Monacelli, 2025).

   This paper: Develop a macro-model in which insurer constraints, housing markets, and consumption are endogenous.

Roadmap

1. Introduction
2. **Empirical analysis**
3. Structural model
4. Conclusion

Data sources

Insurance data

- **California Department of Insurance**:
  - Zip-code×Insurer×Year: insurance exposures and premiums, 2009-2022.
  - Zip-code×Year: number of insurer- and consumer-initiated non-renewals, new policies, coverage limits, fire risk scores, fire losses.
- **National Association of Insurance Commissioners (NAIC)**: insurer balance sheet and P&L data.

Zip-code and county level indicators

- **Zillow** Monthly house price indicators
- **American Community Survey (ACS)**: demographics
- **The Fed**: Household debt statistics

Weather impacts

- **New York Fed**: county-level damages from natural disasters; 1996–2023
- **California fire authority**: Wildfire incidents; 1996–2023

Fact 1: Insurance mitigates house price reductions upon weather shocks.

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Local projection/DiD setup (Dube et al., 2023):

$$Y_{z,t+h} - Y_{z,t-1} = \beta_h \cdot \mathbf{1}_{z,t}^{fire} \cdot \text{High Insurance}_{z,t} + \gamma' X_{c,t} + \eta_z + \tau_t \times \tau_y + \varepsilon_{z,t+h}$$

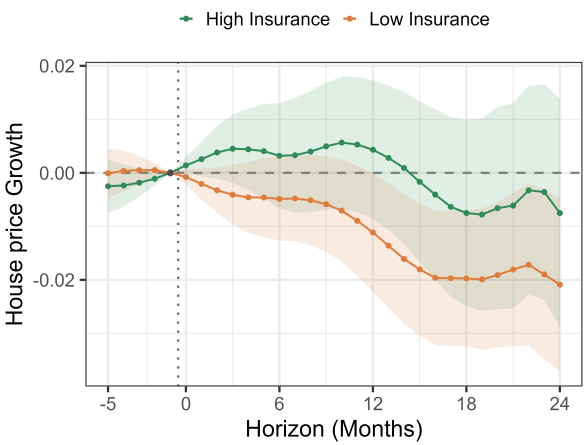
- $Y_{z,t}$: Log average house price in month t in zip-code z (California)
- $\mathbf{1}_{z,t}^{fire}$: Equals 1 if the zip-code experiences a severe wildfire
- $\text{High Insurance}_{z,t}$: Equals 1 if the average ratio of fire coverage to house value is above the median
- Controls: lags of house prices, changes in insurance prices, previous fire indicators.

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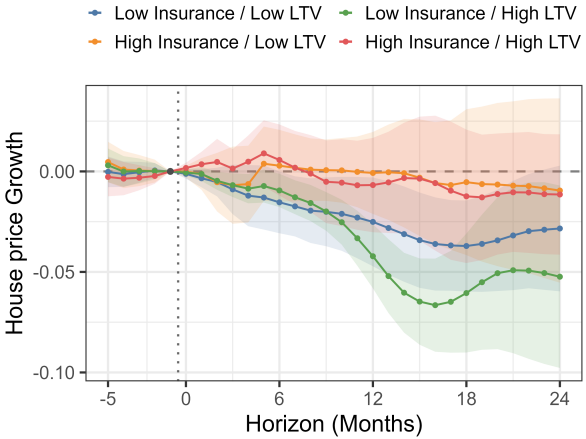
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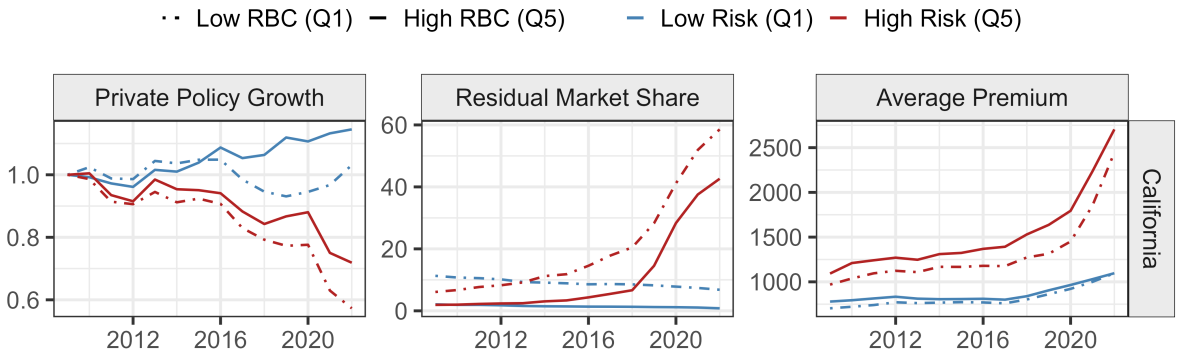
Fact 1: Insurance mitigates house price reductions upon weather shocks, ctd.

$$Y_{z,t+h} - Y_{z,t-1} = \beta_h \cdot \mathbf{1}_{z,t}^{fire} \cdot \text{High Insurance}_{z,t} \cdot \text{High Debt}_{z,t} + \gamma' X_{c,t} + \eta_z + \tau_t \times \tau_y + \varepsilon_{z,t+h}$$

- High $\text{Debt}_{z,t}$: Equals 1 if average outstanding debt-to-house value ratio is above the median



Fact 2: Insurance market contractions steepen in insurer financial constraints



RBC = Risk-Based Capital Ratio

Fact 3: Insurance market contractions negatively affect house prices conditional on weather exposure

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$$Y_{z,t} = \beta M_{z,t} + \gamma' X_{c,t} + \eta_z + \tau_t + \epsilon_{z,t} \tag{1}$$

- $Y_{z,t}$: log median/mean house price
- $M_{z,t}$: (i) log avg. insurance price, (ii) log # insurer-initiated cancellations
- η_z, τ_t: zip-code and year fixed effects
- $X_{c,t}$: county & zip-code controls (demographics, income, housing stock, insurance policy characteristics, disaster damages).

Identification: instrument prices + non-renewals with *out-of-county* weather losses, weighted by insurers' exposure shares, excluding nearby counties ($d > \bar{D}$) to avoid spatial spillovers:

$$Z^{\bar{D}}_{z,t} = \log\left(\sum \underbrace{\omega_{i,z,t-1}}_{\text{market share}} \sum_{d_{c(z),\tilde{c}} > \bar{D}} \underbrace{\omega^{s}_{\tilde{c},i,t-1}}_{\text{portfolio weight}} \text{WeatherLosses}_{\tilde{c},t-1} \right) \tag{2}$$

Fact 3: Insurance market contractions negatively affect house prices conditional on weather exposure, ctd.

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	Log(Median HP)		Log(Mean HP)		Log(Median HP)		Log(Mean HP)	
	OLS	IV	OLS	IV	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log Ins. Price	-0.08^{***}	0.05	-0.10^{***}	-0.08^{**}				
	(0.03)	(0.05)	(0.02)	(0.03)				
Log Non-renewals					-0.02^{***}	-0.08^{**}	-0.002	-0.05^{***}
					(0.006)	(0.04)	(0.003)	(0.02)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Zip-code FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	12,768	12,731	11,753	11,746	7,361	7,361	7,002	7,002
R^2	0.96	0.96	0.99	0.99	0.98	0.98	0.99	0.99
Within R^2	0.08	0.07	0.16	0.15	0.04	0.04	0.17	0.17

Roadmap

1. Introduction
2. Empirical analysis
3. **Structural model**
   - Setup
   - Calibration
   - Impulse Response Functions
   - Planner's problem, taxation and welfare
4. Conclusion

Motivation and overview

- Empirical results show that:
  - insurance protects house prices both directly and indirectly.
  - insurer financial conditions affect insurance contractions.

→ **Intertemporal trade-off** in balancing household and insurer fragility.

- Using a dynamic housing model with insurers, I:
  - Solve for a decentralized equilibrium
  - Derive the optimal and time-consistent insurance allocation
  - Decentralize planner's allocation with taxes

- Insurance subsidy or tax corrects for the failure (of households) to internalize:
  - House price reductions upon weather shocks (in high leverage and low insurance states).
  - The possibility of insurance market crashes.

Setup

- Time is discrete and infinite.
- Representative household chooses:
  - Consumption c_t
  - Housing s_{t+1}
  - Savings b_{t+1}
  - Insurance x_{t+1} as a fraction of possible damage
  - Home equity redraws l_t
- Household maximizes lifetime utility:

$$u(c_t, s_t) = \frac{(c_t + \nu_s g(s_t))^{1-\psi}}{1-\psi} \tag{3}$$

- Insurers, constrained by capital regulation, supply insurance and invest.
- Disasters occur with probability π^d, destroying houses by fraction ϕ, with ϕ time-varying.
- State variables:
  - Insurance
  - Current savings
  - Insurer net worth
  - Exogenous shocks: disaster indicator $\mathbf{1}$, severity ϕ, productivity z, Interest rate R, Collateral constraint tightness κ.

Household constraints

Budget constraint

$$c_t + \frac{b_{t+1}}{R_t} + s_{t+1}p_t^s + p_t^x x_{t+1} s_{t+1} = b_t + w_t + p_t^s s_t + \mathbf{1}_t \phi_t p^r s_t (x_t - 1) + l_t - f(l_t) \tag{4}$$

Collateral constraint

$$-\frac{b_{t+1}}{R_t} \leq \kappa_t \left(p_t^s s_t \left(1 - \mathbf{1}_t \phi_t\right) + \mathbf{1}_t \phi_t s_t x_t p^r - l_t \right) \tag{5}$$

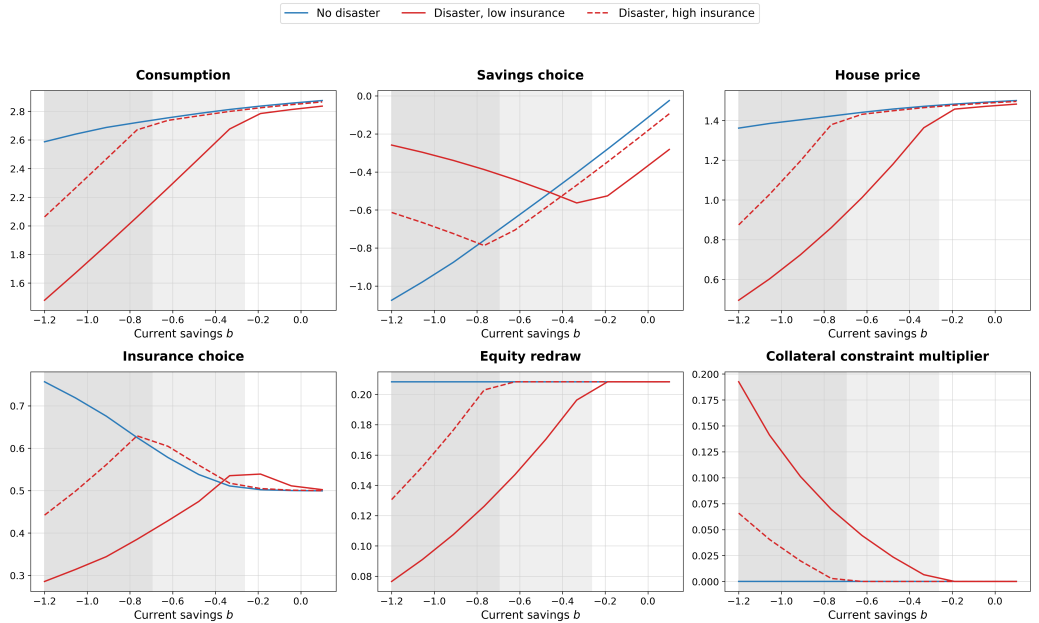
p_t^s	house price
p_t^x	insurance premium per unit of housing
s_{t+1}	house size normalized to 1
x_{t+1}	insurance coverage $\in [0, 1]$
b_{t+1}/R_t	saving at exo. rate R_t (< 0 for borrowing)
w_t	wage;
$\mathbf{1}_t$	disaster indicator ($= 1$ if disaster occurs)
ϕ_t	fraction house destroyed
$\mathbf{1}_t \phi_t p^r s_t x_t$	insurance payout
κ	collateral constraint tightness

Optimality conditions for savings and insurance (decentralized equilibrium)

$$b_{t+1} :: \qquad U'(t) = \beta R_t \mathbb{E}_t \left[U'(t+1) \right] + \underbrace{\mu_t}_{\text{SV collateral con.}} \tag{6}$$

$$x_{t+1} :: \qquad p_t^x = \mathbb{E}_t [\mathbf{1}_{t+1} \phi_{t+1} p^r (\underbrace{m_{t+1}}_{\text{Stochastic Discount Factor}} + \underbrace{\frac{\mu_{t+1} \kappa_{t+1}}{\lambda_t}}_{\text{Collateral Value of Insurance}})] \tag{7}$$

Household policy functions



Insurer problem

Insurers are risk-neutral, accumulate net worth n_t, and choose insurance x_{t+1}, assets a_t and dividend payouts δ_t to maximize the PV of dividends J_t.

$$\begin{aligned} \text{First stage (dividend payouts):} \quad & J(n_t, \theta_t) = \max_{\delta_t} \{\delta_t n_t - f(\delta_t) + W(N_t, \theta_t)\} \\ \text{Second stage (insurance supply):} \quad & W(N_t, \theta_t) = \max_{x_{t+1} \geq 0} \beta \, \mathbb{E}_{\mathcal{S}_{t+1}|\mathcal{S}_t} \left[J(n_{t+1}, \theta_{t+1}) \right] \end{aligned} \tag{8}$$

subject to

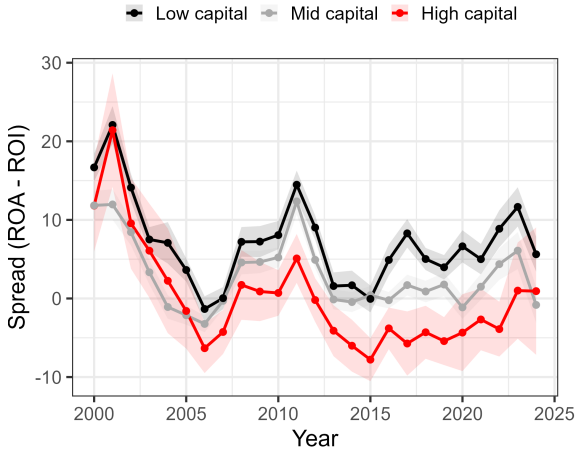
$$\begin{aligned} \text{Net worth post-dividends:} \quad & N_t = (1 - \delta_t) n_t \\ \text{Investable assets:} \quad & a_t = N_t + p_t^x x_{t+1} s_{t+1} \\ \text{Net worth transition:} \quad & n_{t+1} = a_t (1 + R_{t+1}^k) - \mathbf{1}_{t+1} \phi_{t+1} x_{t+1} p^r - \chi_1 x_{t+1}^{\chi_2} \\ \text{Capital regulation:} \quad & \zeta x_{t+1} p^r \leq W(N_t, \theta_t) \end{aligned}$$

Final good firms close the model and determine equilibrium return on assets and wages.

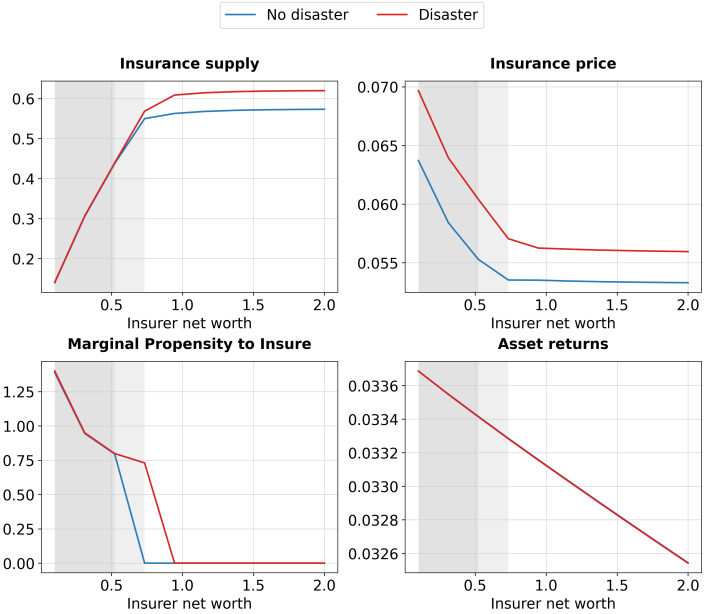
Insurer problem - prediction

- Capital regulation drives a wedge between Return on Assets (ROA) and Return on Insurance (ROI), and the wedge increases in the severity of financial constraints.

Spread Return on Assets and Return on Insurance (U.S. P&C insurers)



Insurer policy functions



Calibration and Moments

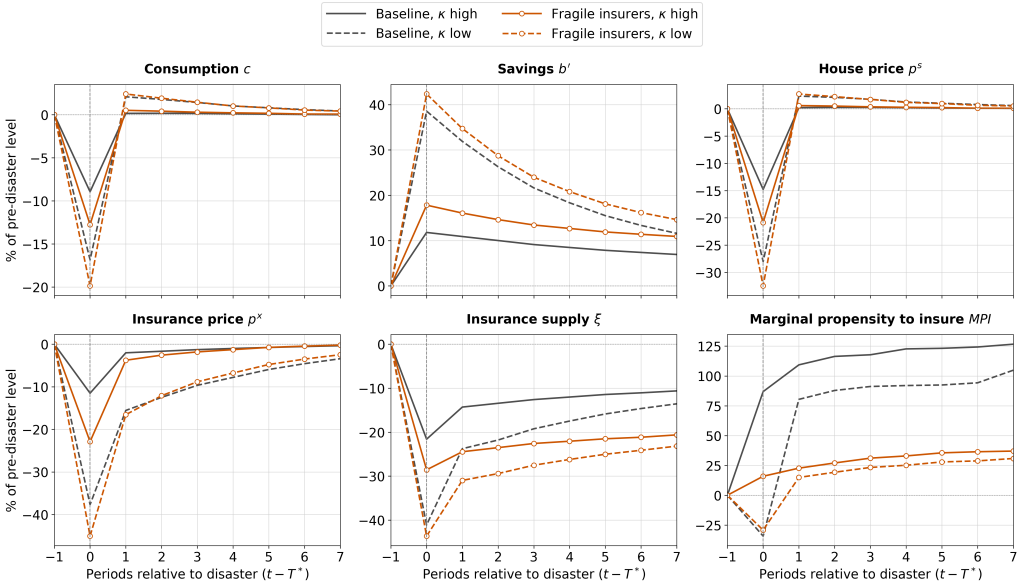
Calibration targets

Moment	Model	Target / Data
Targeted moments		
Insurance gap	42.9%	45%
Return on invested assets	3.3%	3.2%
Insurers' share of corporate bonds	2.8%	3.3%
Marginal propensity to insure	0.24	0.50
Constrained insurers	28.9%	27.6%
Housing cost / total expenditures	0.34	0.33
Rebuilding cost / home value	0.63	0.63
Home equity redraw / home value	0.14	0.14
Output volatility σ_y	0.001	0.01
Untargeted moments		
Housing damages / GDP	0.63%	0.60%
ROA–ROI spread	2.2%	3.0%
Premium income / net worth	4.0%	3.8%
Dividend payouts / net worth	4.5%	8.2%

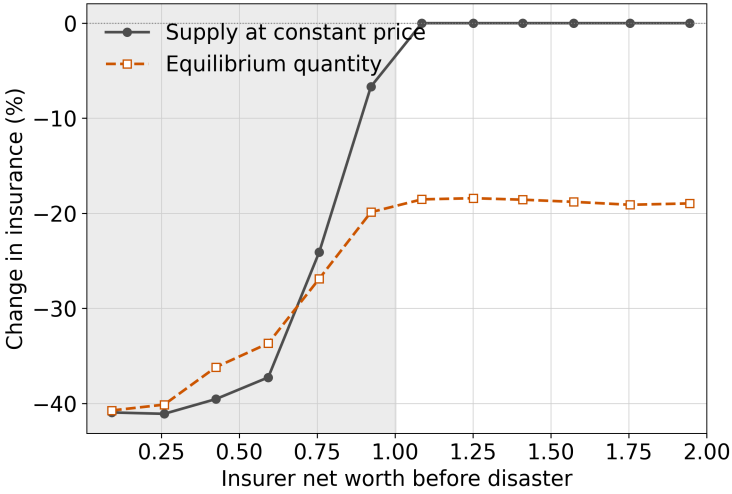
Key parameters

Parameter	Value
Disasters	
Disaster probability π^d	0.0523
Severity volatility σ_θ	0.9
Households	
Risk aversion ψ	2
Discount factor β	0.97
Housing weight ν_s	0.041
Collateral states κ^H, κ^L	0.95, 0.75
Rebuilding cost p^r	0.81
Redraw cost ϑ_l	4.5
Insurers	
Capital constraint ζ	0.95
Linear expenses χ_1	0.031
Quadratic expenses χ_2	2

DE - Impulse Response Functions - disaster shock



Impulse Response Function for insurance quantities by pre-disaster net worth



Planner's Problem

- Planner chooses debt and insurance on behalf of the HH, correcting two externalities:
  - House price crash risk.
  - Insurance market crashes.
- Adopt a primal approach to optimal policy analysis (Bianchi and Mendoza, 2018), where the planner maximizes HH welfare subject to:
  - Budget constraint
  - Collateral constraint
  - Implementability constraint (household FOC for housing)
  - Market clearing constraint

Optimality Conditions (Planner's Equilibrium)

$$c_t :: \quad \lambda^* = u'(c,1) - u''(c,1)\xi^* p^s, \tag{9}$$

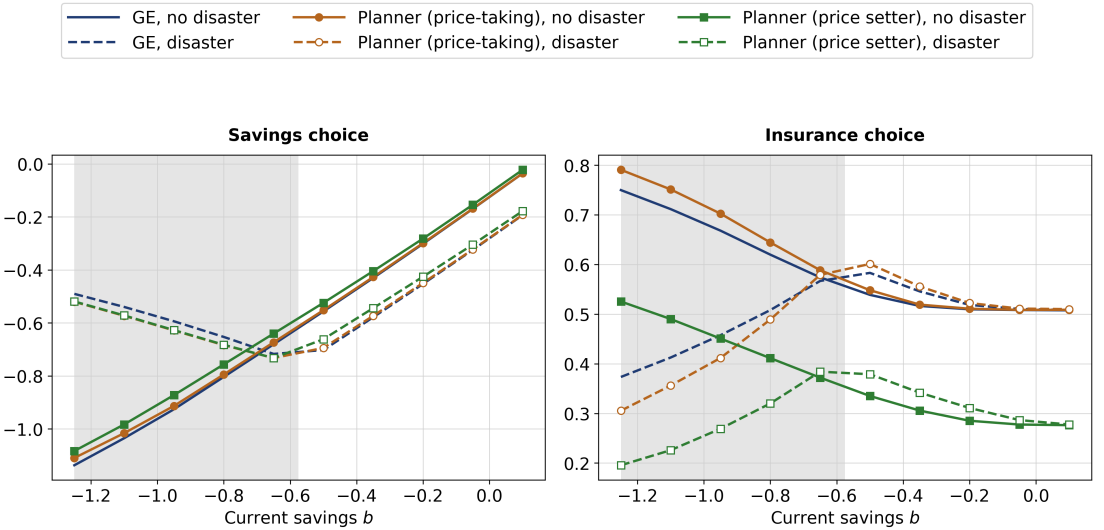
$$b_{t+1} :: \quad \lambda^* = \beta R \, \mathbb{E}_t \left[\lambda'^* + \xi^* \widetilde{\Omega}'_{b'} \right] + \mu^*, \tag{10}$$

$$x_{t+1} :: \quad \lambda^* p^x + \Psi^* = \beta \, \mathbb{E}_t \left[(\lambda'^* + \mu'^* \kappa') \mathbf{1}' \phi' p^r + V_{N'} \Lambda^{N'}_{x'} + \xi^* \widetilde{\Omega}'_{x'} \right]. \tag{11}$$

$$V_N = \Psi^* \underbrace{\mathcal{X}_N(p^x, n, \mathcal{S})}_{\text{MPI}} + \beta \, \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[\Lambda^{N'}_N \left(V_{N'} + \xi^* \tilde{\Omega}'_{N'} \right) \right] \tag{12}$$

- λ^*: Shadow value of consumption, which accounts for how changes in marginal utility affect house prices and relax the implementability constraint.
- μ^*: Social value of relaxing the borrowing constraint
- Ψ^*: Component of insurance prices reflecting price impact through intermediary constraints.
- $V_{N'} \Lambda^{N'}_{x'}$: Marginal continuation value of next-period net worth times how insurance marginally affects future net worth.
- $\widetilde{\Omega}'$: Anticipation of future planners' choices.

Planner's policy functions



The planner's insurance allocation can be decentralized with a tax

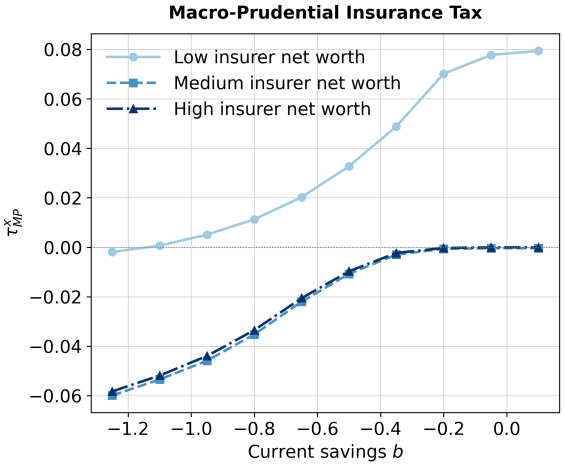
The planner's allocation can be decentralized with an insurance tax as the ratio of prices:

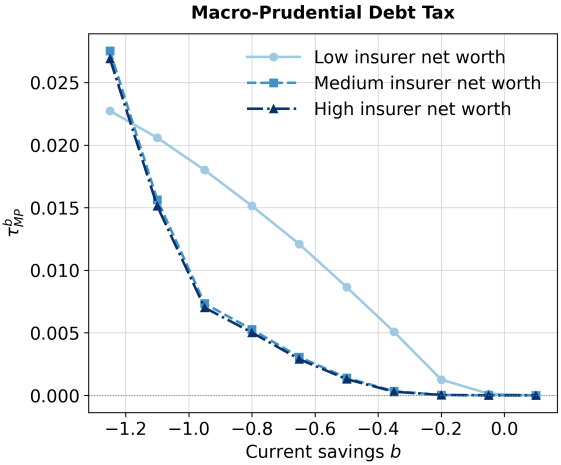
$$\tau_t^x = \frac{p_{t,\mathrm{DE}}^x}{p_{t,\mathrm{PL}}^x} - 1.$$

Macro-prudential (MP) tax is levied in good times (when today's constraints do not bind)

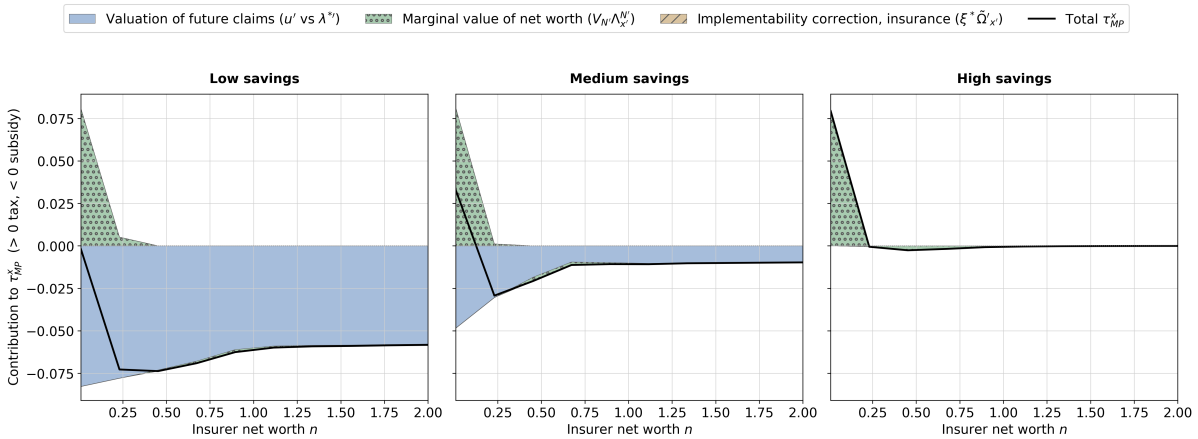
$$\tau_{t,MP}^x = \frac{\mathbb{E}_t\left[\left(u'(c',1) + \mu'\kappa'\right)\mathbf{1}'\phi' p^r\right]}{\mathbb{E}_t\left[(\lambda'^* + \mu'^*\kappa')\mathbf{1}'\phi' p^r + V_{N'}\Lambda_{x'}^{N'}\right]} - 1$$

Macro-prudential insurance tax and debt tax





Decomposition of macro-prudential tax/subsidy (no disaster state)



- Subsidizing insurance is optimal, but the subsidy is smaller when insurer net worth is low.

Planner simulation and welfare analysis

	Decentralized equilibrium		Planner (price-taking, $\Psi^* = 0$)		Planner (price-setting, $\Psi^* \neq 0$)	
	Mean	SD	Mean	SD	Mean	SD
Summary statistics						
House price p^s	1.527	0.057	1.528	0.042	1.527	0.057
Consumption	2.955	0.065	2.956	0.049	2.965	0.069
Marginal Propensity to Insure	0.332	0.364	0.256	0.347	0.639	0.034
Insurance gap	43.6%	–	41.2%	–	99.2%	–
Insurer constraint binds	39.6%	–	29.9%	–	99.7%	–
House price drop at disaster	16.3%	–	11.5%	–	15.6%	–
Insurer net worth drop at disaster	25.6%	–	24.8%	–	0.2%	–
Taxes and welfare						
MP Debt tax τ^b	–	–	0.47%	0.06%	0.56%	0.10%
MP Insurance tax τ^x	–	–	−2.65%	1.89%	−2.45%	0.73%
ρ_{τ^b,τ^x}	–	–	0.334	–	0.077	–
Welfare gain	–	–	0.19%	–	−2.82%	–

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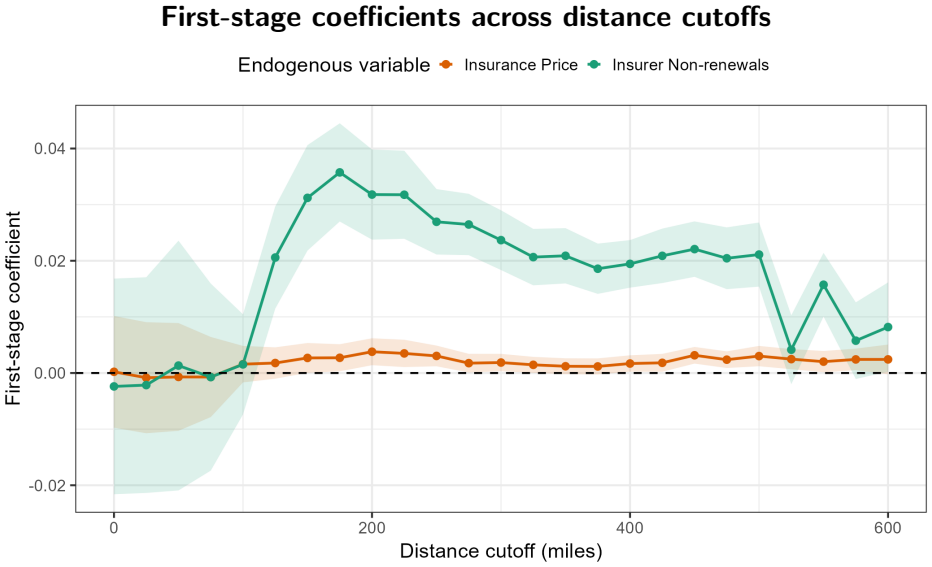
Conclusion

- This paper develops a dynamic housing model with disaster risk in which intermediary constraints are endogenous.
- Model (and empirics) reveal a basic **intertemporal trade-off** between household and insurer fragility.
- Comparing a social planner's allocation with the decentralized equilibrium yields policy implications:
  - Subsidizing insurance is welfare-enhancing, but the optimal policy is dynamic / state-dependent.
  - Regulators directly influencing insurance prices lowers welfare as insurance gaps become persistent.

Appendix - Fact 3: Insurance market contractions negatively affect house prices conditional on weather exposure, ctd.

	Log Ins. Price (1)	Log Ins. Non-renewals (2)	Log Ins. Price (3)	Log Ins. Non-renewals (4)	Δ Log RBC (avg) (5)	Δ Log Net Worth (avg) (6)
Z^{Resid}	0.63***		2.0***			
	(0.09)		(0.77)			
Z^{200}		0.03***		0.23**	−0.01***	−0.003***
		(0.004)		(0.09)	(0.001)	(0.0004)
Log Net Worth			0.06***	0.14		
			(0.01)	(0.09)		
$Z^{Resid} \times$ Log Net Worth			−0.10*			
			(0.06)			
$Z^{200} \times$ Log Net Worth				−0.01**		
				(0.006)		
Zip-code FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
F-test	52.2	51.7	153.3	81.9	38.5	27.1
Observations	14,469	8,029	14,469	8,029	14,935	15,046
R^2	0.92	0.96	0.93	0.96	0.29	0.71
Within R^2	0.23	0.05	0.29	0.05	0.02	0.02

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House prices, insurer net worth and welfare

